



Gradence

DASHBOARDS, BI TOOLS & AI IN ANALYTICS

Week 4

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Key Performance Indicators (KPI's)

INTRODUCTION TO BUSINESS INTELLIGENCE (BI)

What is Business Intelligence?

BI = process of transforming data into actionable insights

Purpose

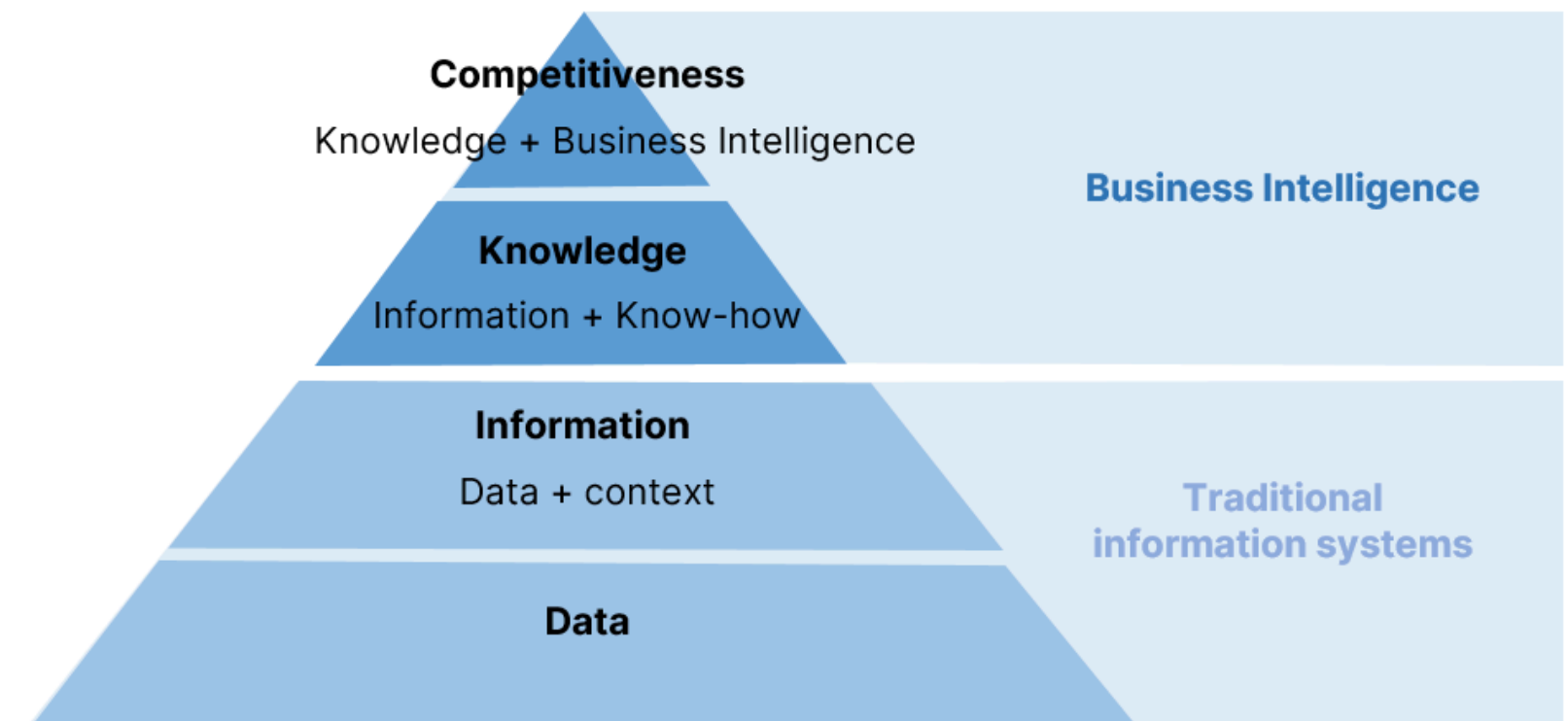
Enable data-driven decision making across all organisational levels

Core Function

Bridge the gap between IT systems and business strategy

Key Outcome

Turn information into competitive advantage



FROM DATA TO DECISIONS

Raw Data	Key Performance Indicators	Strategic Decisions
Customer transaction records	Monthly Revenue: £2.3M	Increase marketing spend in high-performing regions
Website clickstream data	Customer Acquisition Cost: £45	Optimise product pricing strategy
Sales performance metrics	Conversion Rate: 3.2%	Expand successful product lines
Marketing campaign results	Average Order Value: £85	Improve customer retention programmes

POPULAR BI TOOLS OVERVIEW

Choosing the Right Tool:

- Budget and licensing requirements
- Technical expertise of your team
- Integration with existing systems
- Scalability and performance needs

Enterprise Leaders



Industry standard for data visualization



Microsoft Power BI

Seamless Office 365 integration



Associative analytics engine



SQL-based modelling approach

Modern Cloud Solutions



Open-source, user-friendly interface



Google Data Studio

Free, web-based reporting

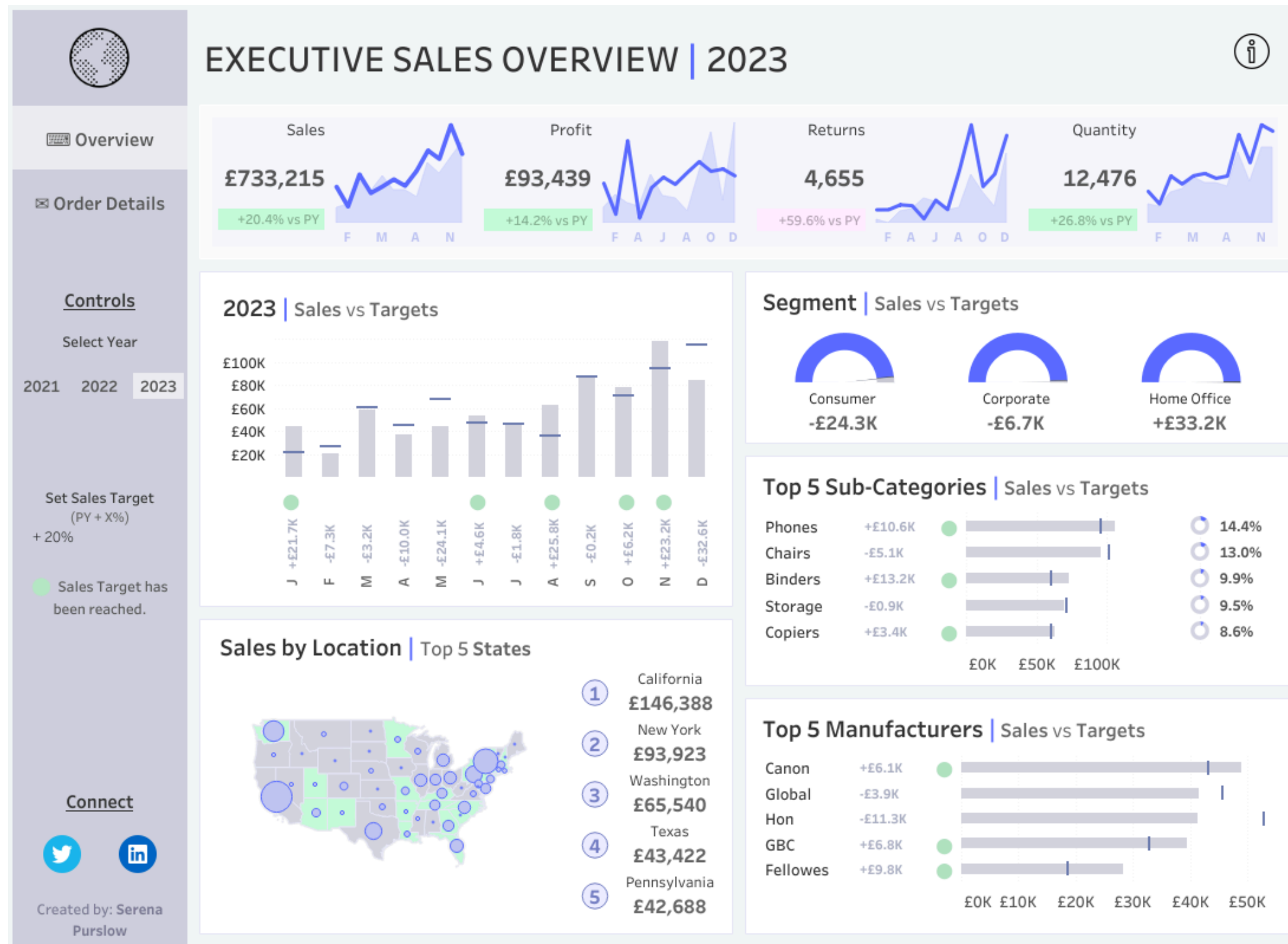
TABLEAU VS POWER BI



Features		
Data Virtualization	✓	✓
Data Connectivity	✓	✓
Data Querying	✓	✓
Data Management - ETL	✓	✓
Data Security	✓	✓
Natural Language Processing	✓	✓
Ease of Learning	✓	✓
Free Version	Power BI Desktop	Tableau Public

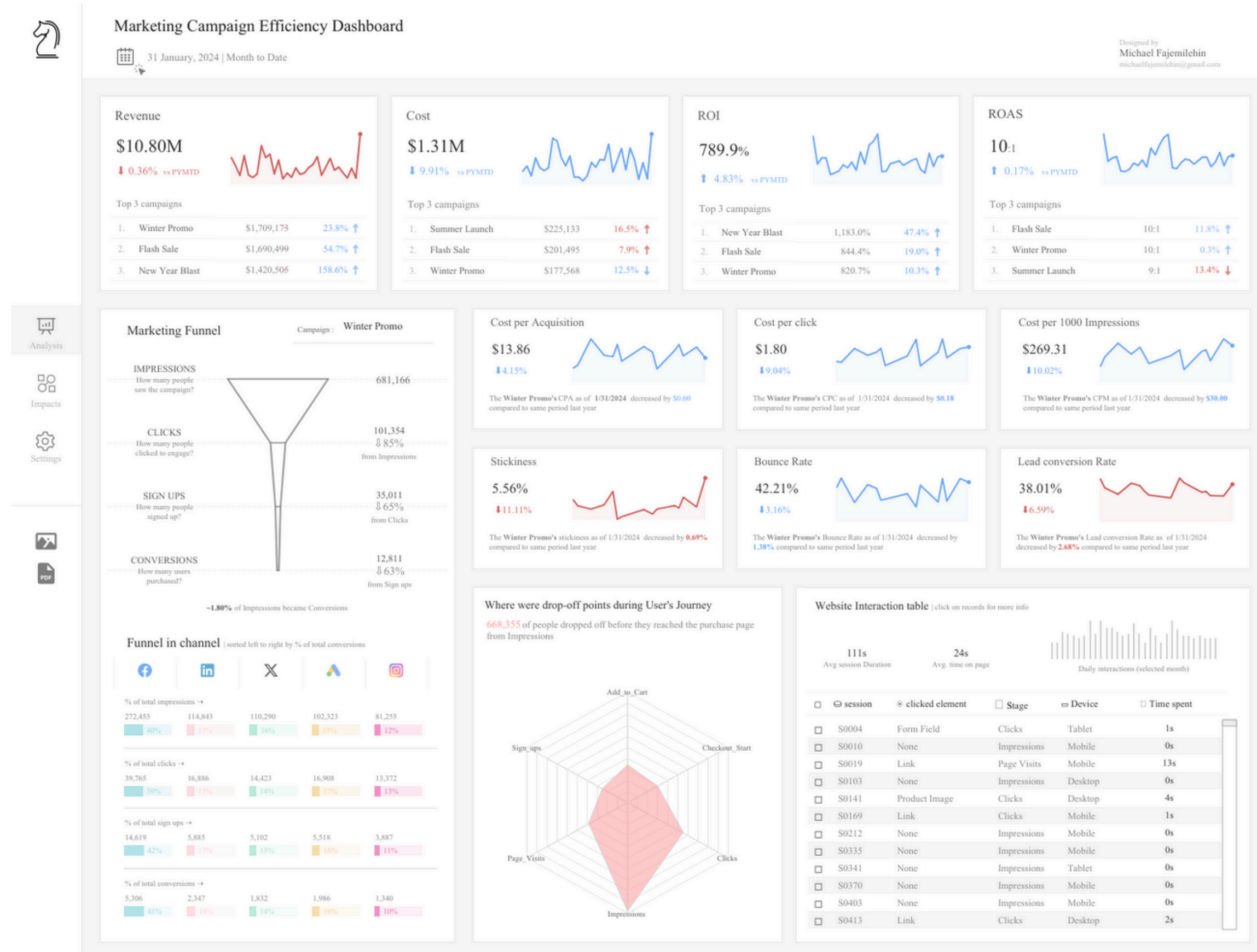
✓ Good ✓ Best

SALES PERFORMANCE DASHBOARD



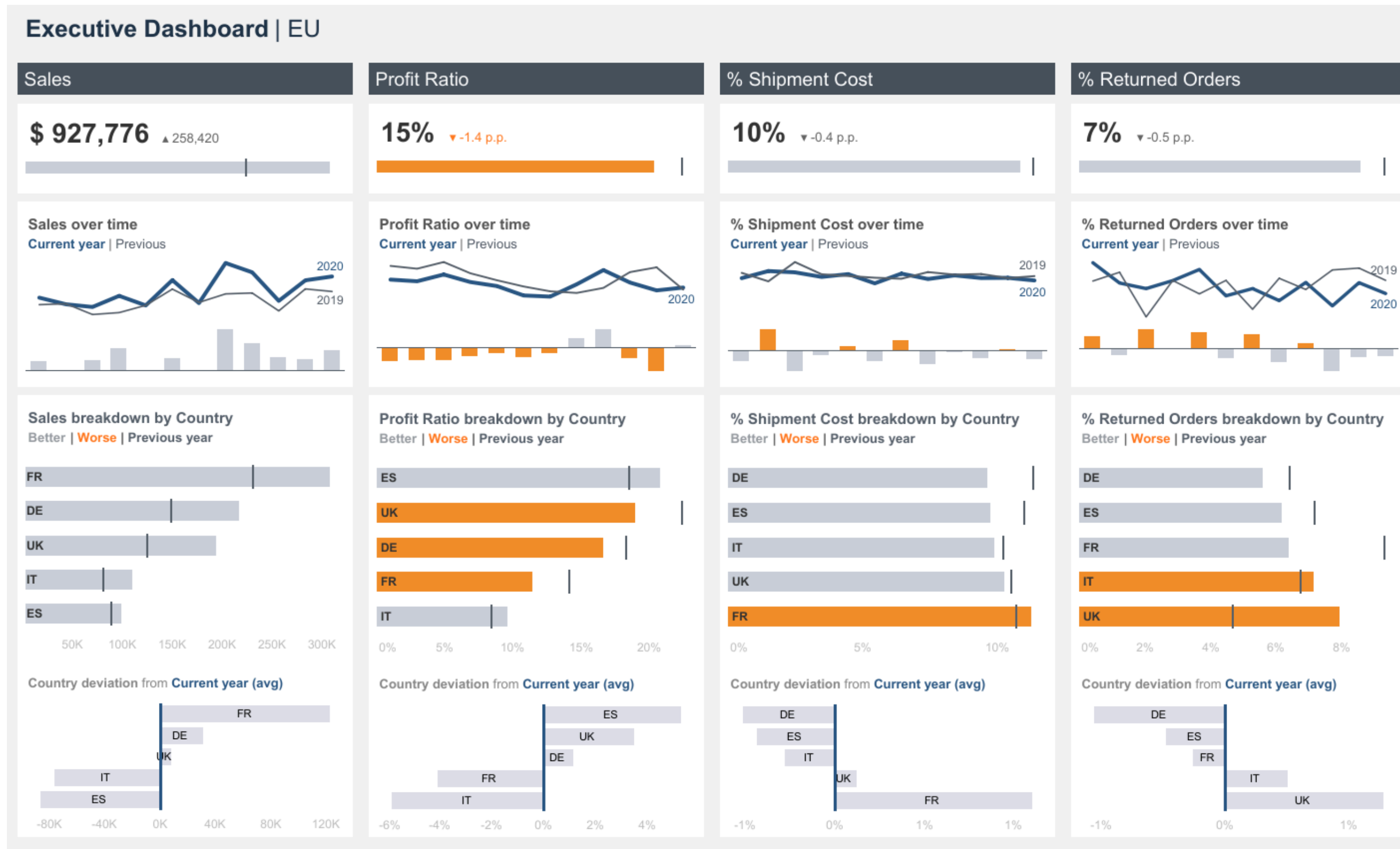
- Tracks overall revenue performance across time and regions
- Monitors sales targets vs actuals to measure success
- Identifies top-performing products, segments, or regions
- Helps spot sales trends, seasonality, and anomalies
- Supports sales forecasting and strategic planning
- Enables quick performance comparison by country, product, or team

MARKETING CAMPAIGN ROI



- Measures campaign efficiency and ROI in real time
- Tracks cost metrics like CPA, CPC, CPM for budgeting
- Analyses user funnel performance (impressions → conversions)
- Identifies top-performing channels and campaigns
- Highlights drop-off points in the user journey
- Provides insights for campaign optimisation and reallocation

EXECUTIVE SUMMARY



- Provides a high-level overview of business performance
- Combines key metrics: sales, profit, costs, returns
- Enables leadership to monitor performance across countries
- Identifies operational inefficiencies or cost drivers
- Supports strategic decision-making with comparative insights
- Highlights areas needing attention without deep drilling

TABLEAU PUBLIC: YOUR GATEWAY TO DATA VISUALISATION

- Full Tableau Desktop functionality for free
- Access to comprehensive visualisation library
- Drag-and-drop interface for easy chart creation
- Community of millions of public visualisations

Perfect For:

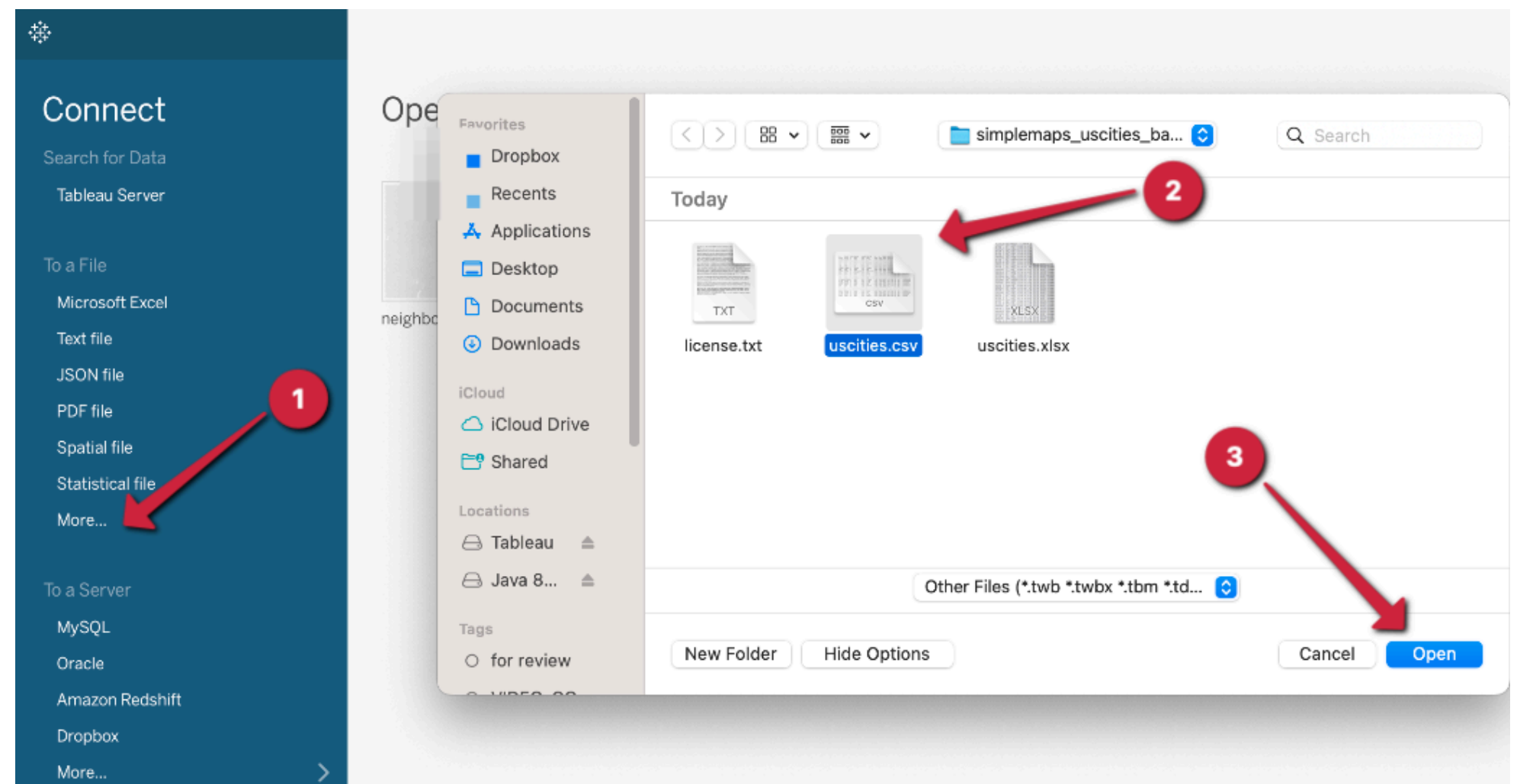
- Learning data visualisation skills
- Building a public portfolio
- Sharing insights with broader audiences
- Academic and educational projects

 Tableau Desktop All listed data sources can be connected to using this App.	 Tableau Public Only Excel and text files can be connected to using this App.	 Tableau Reader Does not need to connect to a Data Source.
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GETTING STARTED WITH TABLEAU PUBLIC

Step 1: Upload Your Data

- Supported formats: CSV, Excel, Google Sheets
- Maximum file size: 1GB
- Automatic data type detection
- Preview your data before proceeding



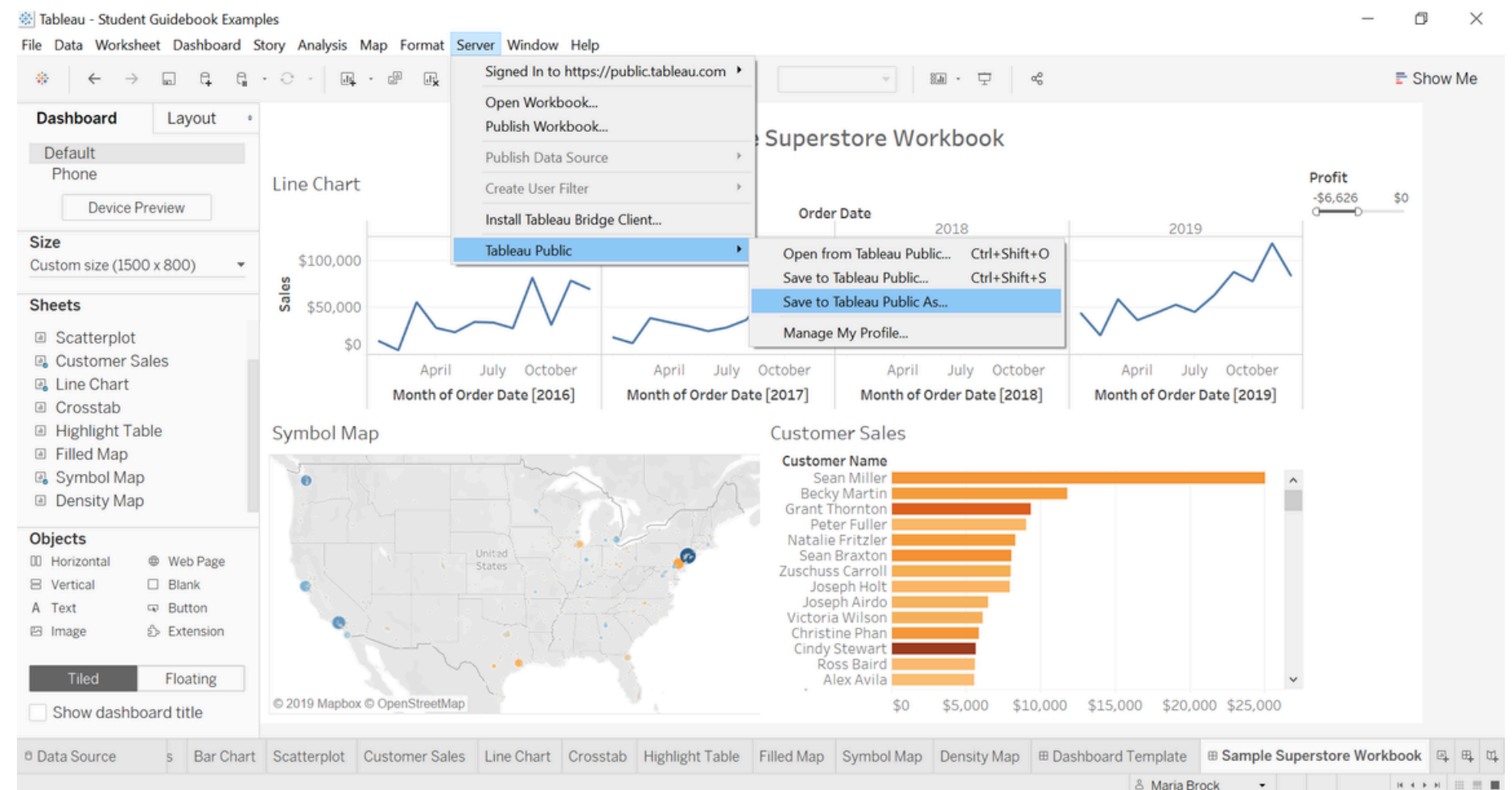
GETTING STARTED WITH TABLEAU PUBLIC

Step 2: Create Your First Chart

- Drag dimensions to columns/rows
- Drop measures onto the view
- Tableau automatically suggests chart types
- Customise colours, labels, and formatting

Step 3: Publish and Share

- Save your work to Tableau Public
- Generate shareable links
- Embed in websites and presentations
- Join the Tableau Public community



ESSENTIAL CHART TYPES



Bar Charts

Compare categories and rankings



Line Charts

Show trends over time



Scatter Plots

Explore relationships between variables



Maps

Visualise geographic data patterns

ADDING INTERACTIVITY

Filters

Allow users to focus on specific data subsets

Parameters

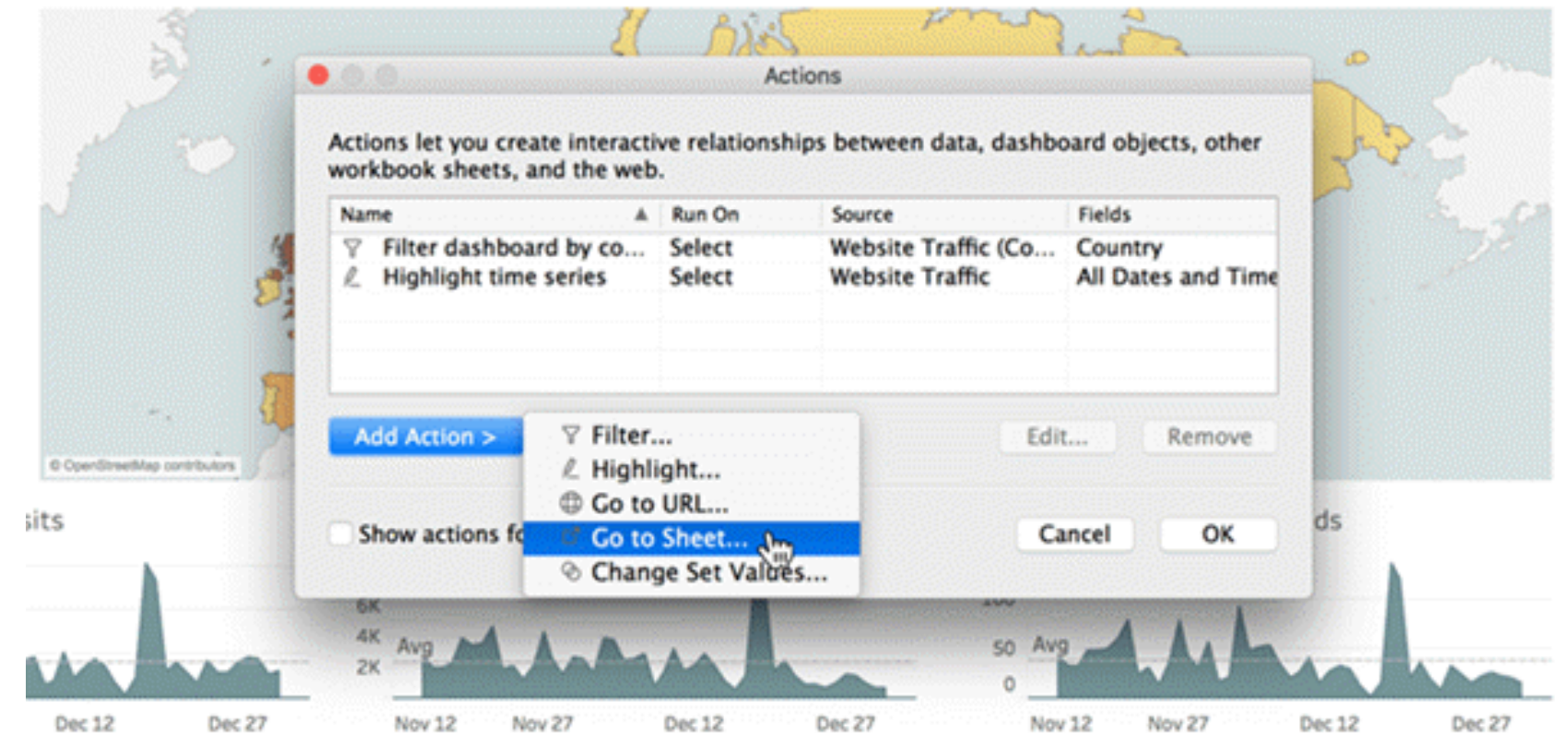
Enable dynamic calculations and scenarios

Actions

Link multiple visualisations together

Tooltips

Provide additional context on hover

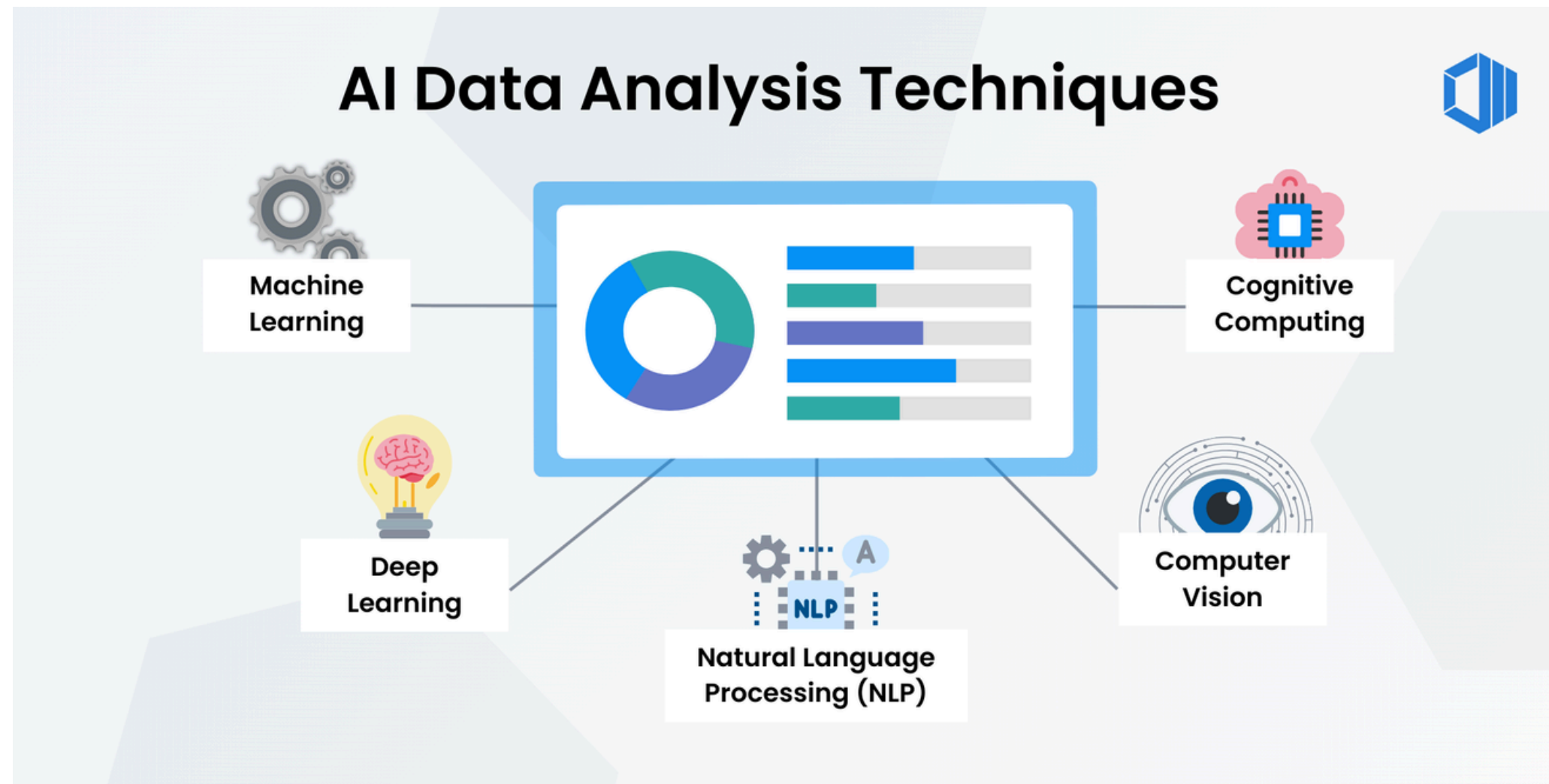


Best Practices:

- Choose charts that match your data and question
- Use colour purposefully to highlight insights
- Keep interactions intuitive and responsive
- Test your dashboard with actual users

AI IN DATA ANALYSIS

The incorporation of artificial intelligence technologies into data analysis workflows to automate, enhance, and accelerate the process of extracting meaningful insights from data.





AI analytics

VS



Traditional analytics

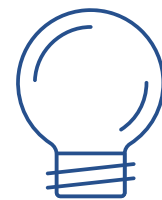
- Automated data preprocessing
- Dynamic pattern discovery
- Real-time insights
- Automated hypothesis generation
- Infinite scalability potential

- Manual data cleaning and preparation
- Rule-based analysis
- Static reporting
- Human-driven hypothesis testing
- Limited scalability

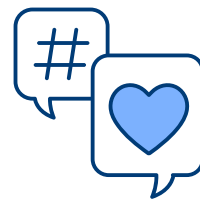
What AI Brings to Analytics



Pattern Recognition – Detect complex trends beyond human analysis



Automation – Speeds up model creation and testing



Natural Language Processing (NLP) – Analyses text data (e.g., reviews, feedback)



Recommendation Systems – Personalised outputs and insights

CORE AI TECHNOLOGIES IN DATA ANALYSIS



Machine Learning

- Supervised learning for prediction
- Unsupervised learning for pattern discovery
- Reinforcement learning for optimization



Natural Language Processing

- Text mining and sentiment analysis
- Automated report generation
- Query processing in natural language



Computer Vision

- Image and video data analysis
- Document processing and OCR
- Visual pattern recognition



KEY BENEFITS OF AI INTEGRATION



Efficiency Gains

- 80% reduction in data preparation time
- Automated anomaly detection
- Real-time processing capabilities



Enhanced Accuracy

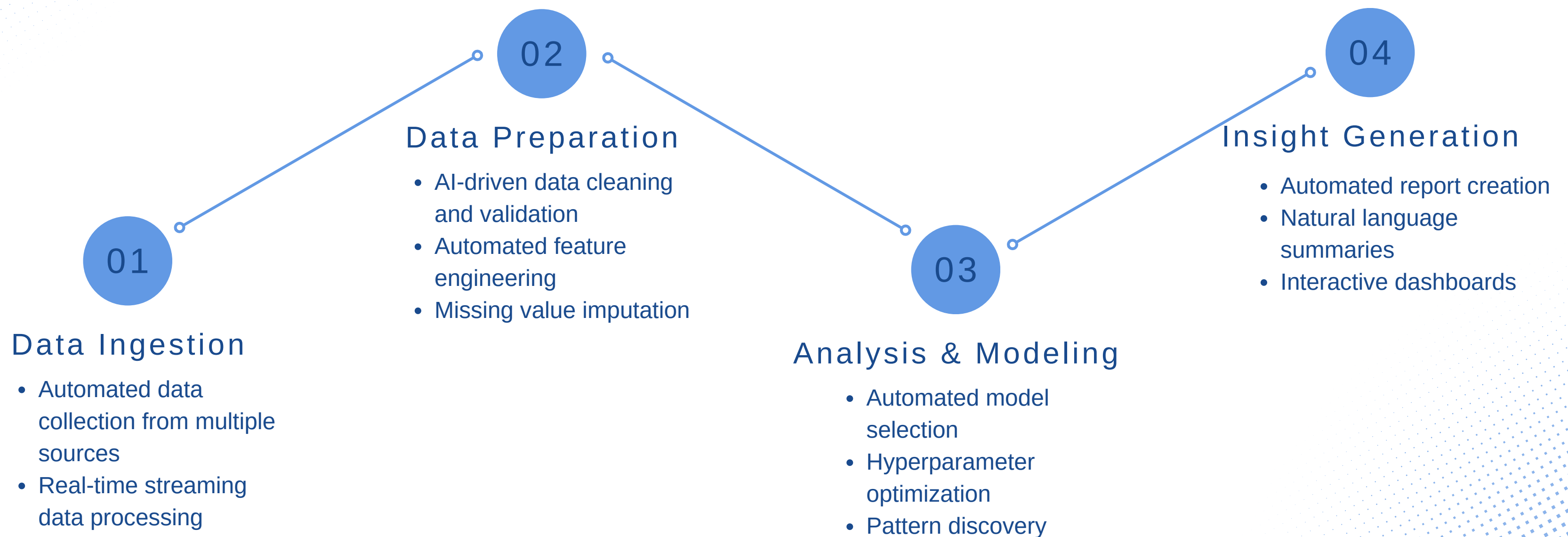
- Reduced human error
- Advanced pattern recognition
- Predictive capabilities



Scalability

- Handle massive datasets
- Process multiple data sources simultaneously
- Adapt to growing data volumes

AI-POWERED DATA ANALYSIS WORKFLOW



WHERE AI FITS IN THE ANALYTICS LIFECYCLE

Data Cleaning & Preparation

AI-powered tools like *Tableau Prep*, *DataRobot*

Insight Generation

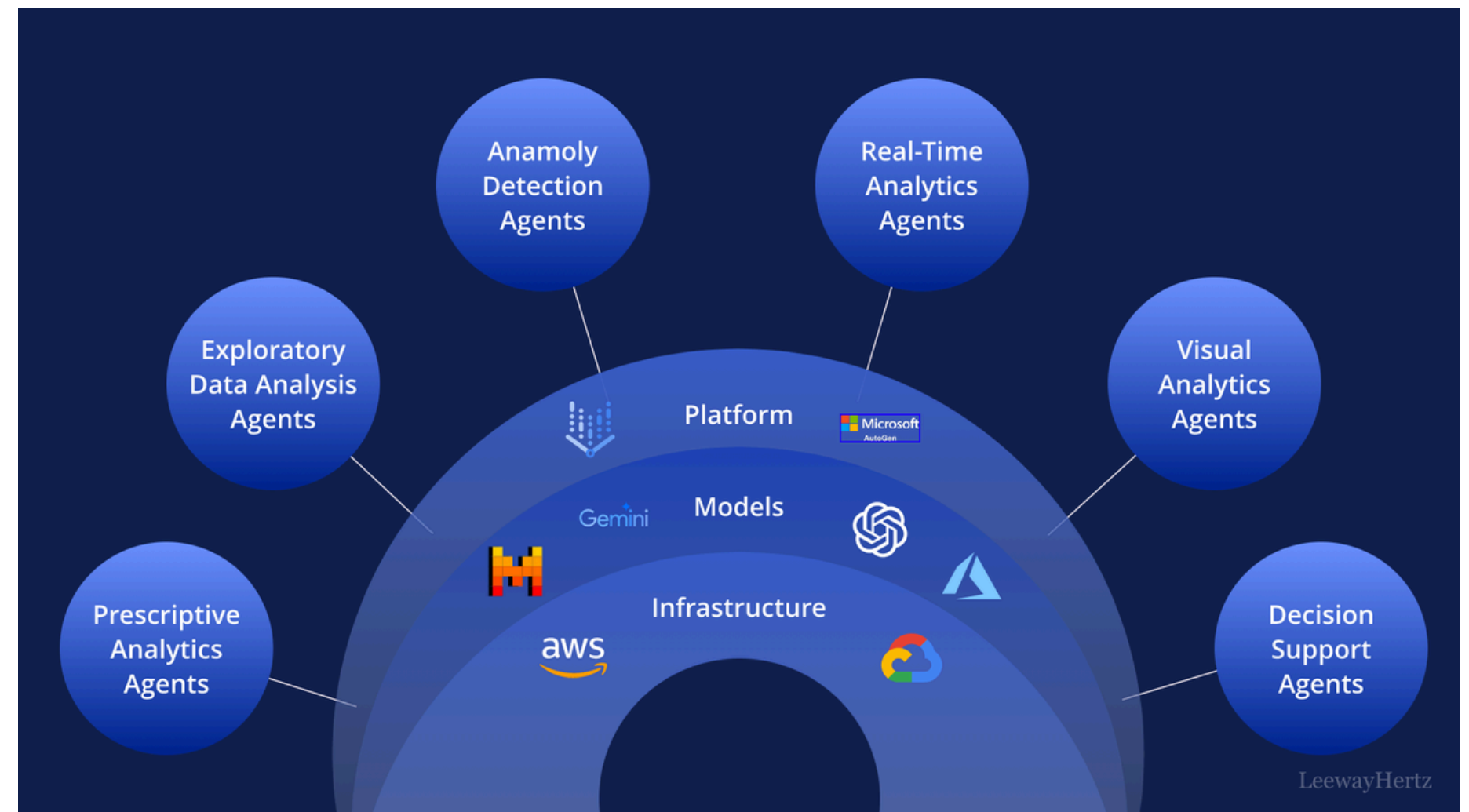
Smart suggestions, anomaly detection

Modelling

AutoML platforms build predictive models

Communication

Generative AI summarises findings or explains visuals



POPULAR AI TOOLS AND PLATFORMS



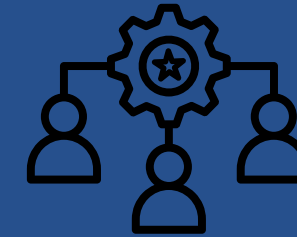
Commercial Platforms

- Microsoft Power BI with AI features
- Tableau with Einstein Analytics
- IBM Watson Analytics
- Google Cloud AI Platform



Open Source Solutions

- Python libraries (scikit-learn, TensorFlow, PyTorch)
- R packages for statistical analysis
- Apache Spark for big data processing
- Jupyter notebooks for interactive analysis



AutoML Platforms

- Google AutoML
- Amazon SageMaker
- H2O.ai
- DataRobot

AI SUCCESS STORY 1 - FRAUD DETECTION

- **Challenge:** Manual review of 100,000+ daily transactions
- **Solution:** Pattern recognition using transaction history and behaviour
- **Results:** 90% reduction in false positives, £2M annual savings
- **Impact:** Improved customer experience, reduced operational costs



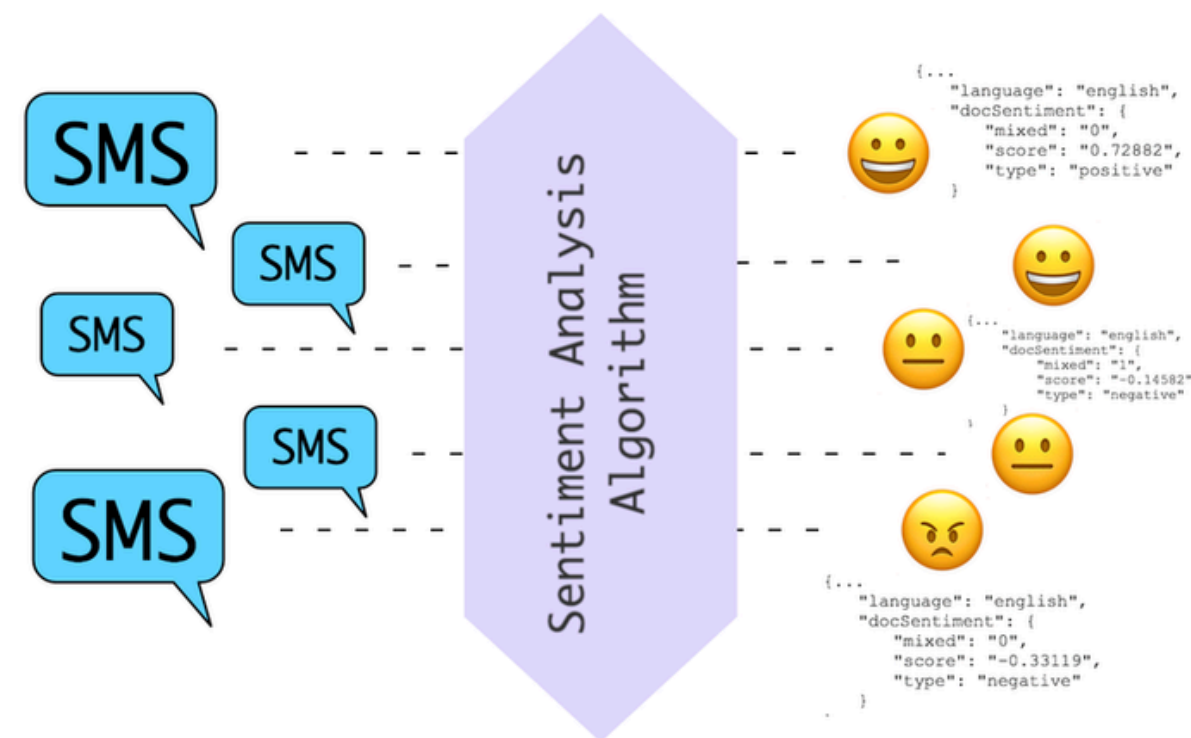
AI SUCCESS STORY 2 - DEMAND FORECASTING

- **Challenge:** Inventory shortages and overstocking across 500+ stores
- **Solution:** Time series ML with weather, promotions, and seasonal data
- **Results:** 15% reduction in inventory costs, 25% improvement in stock availability
- **Impact:** Optimised supply chain, improved customer satisfaction



AI SUCCESS STORY 3 - CUSTOMER SENTIMENT ANALYSIS

- **Challenge:** Manual monitoring of social media and review sites
- **Solution:** NLP analysis of customer feedback across all channels
- **Results:** Real-time brand sentiment tracking, 40% faster response times
- **Impact:** Proactive customer service, improved brand reputation



AI RISKS & ETHICAL CONSIDERATIONS

Technical Risks

Overfitting

Models may not generalise to new data

Data Quality

Poor inputs lead to unreliable outputs

Black Box Problem

Difficulty explaining AI decisions

Bias Amplification

AI can perpetuate historical prejudices

Business Risks

Over-reliance

Reduced human judgement and oversight

Privacy Concerns

Sensitive data exposure and misuse

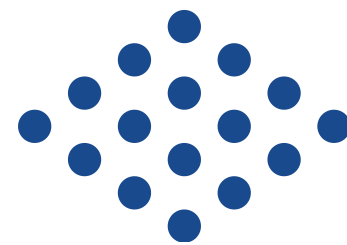
Regulatory Compliance

Evolving legal requirements

Competitive Disadvantage

Falling behind in AI adoption

BEST PRACTICES FOR IMPLEMENTATION

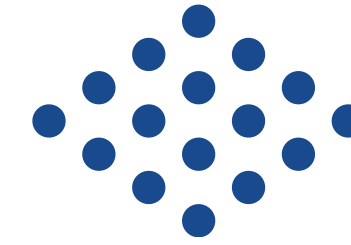


Data Foundation

- Ensure high-quality, clean data
- Establish robust data governance
- Implement proper data security measures

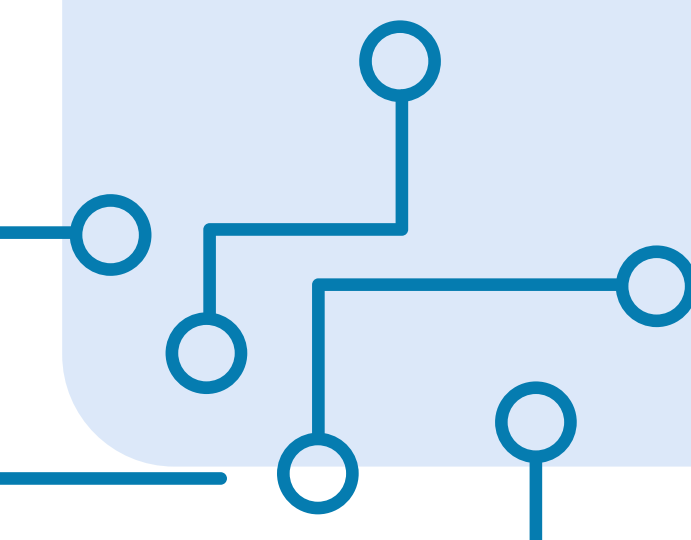
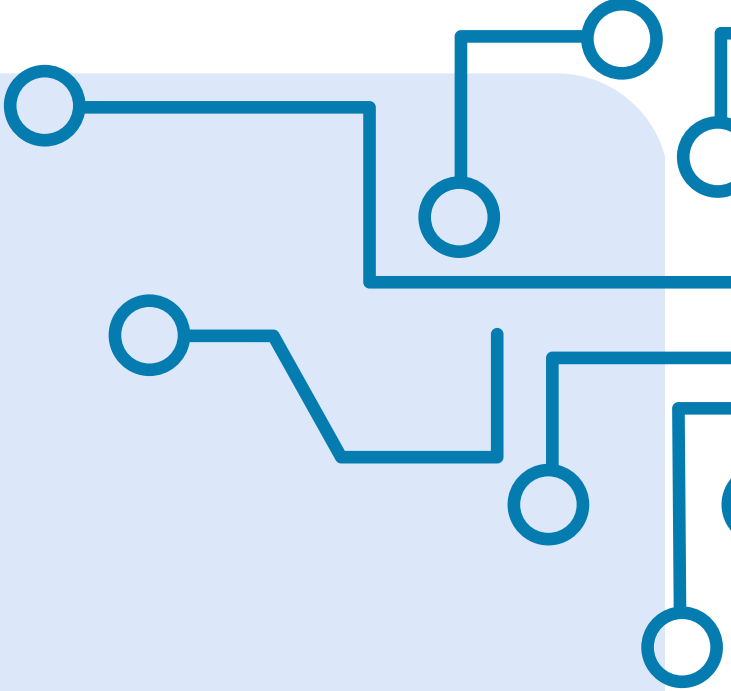
Model Development

- Start with simple models and iterate
- Focus on interpretability and explainability
- Validate models thoroughly
- Monitor performance continuously



Organizational Readiness

- Invest in team training and development
- Establish clear governance frameworks
- Create cross-functional collaboration
- Define success metrics and KPIs



DASHBOARD CREATION WITH AI TOOLS IN ACTION

E-COMMERCE DATASET OVERVIEW

Order_ID	Order_Date	Customer_ID	Product_Name	Product_Category	Revenue (£)	Quantity	Profit_Margin (%)	Customer_Channel	Region	City
100001	06-04-2024	C1456	Reiciendis Plus	Electronics	56.53	2	25%	Referral	South East	Southampton
100002	29-07-2023	C2728	Dolores Pro	Electronics	27.8	2	20%	Social Media	East Midlands	Derby
100003	22-03-2024	C3232	Inventore Lite	Toys & Games	95.35	3	20%	Email	North East	Sunderland
100004	30-10-2024	C2138	Dolorum Lite	Accessories	191.87	3	25%	SEO	North East	Newcastle
100005	30-07-2024	C2470	Ipsum X	Kitchenware	60.26	1	60%	Social Media	South East	Reading
100006	03-12-2024	C1322	Blanditiis X	Home & Living	178.24	3	35%	Referral	South East	Brighton
100007	19-01-2024	C1933	Vitae Pro	Home & Living	172.51	1	50%	Paid Ads	Scotland	Aberdeen
100008	13-06-2025	C2494	Dolore Plus	Accessories	77.5	3	25%	Social Media	South West	Bath
100009	24-10-2023	C2002	Inventore Max	Accessories	82.1	5	35%	Referral	North West	Manchester

Dataset Structure:

- **Orders:** 10,000 transactions over 2 years
- **Products:** 500 items across 8 categories
- **Customers:** 2,500 unique buyers
- **Geography:** UK regions and cities

Business Questions We'll Answer:

- Which products drive the most revenue?
- How do sales vary by season and region?
- What's our customer acquisition cost by channel?
- Where should we focus our marketing efforts?

DATA MODELLING IN TABLEAU

CALCULATED FIELDS:

- **Profit Margin** = (Revenue - Cost) / Revenue
- **Customer Lifetime Value** = SUM(Revenue) / COUNT(DISTINCT Customer ID)
- **Month-over-Month Growth** = (Current Month - Previous Month) / Previous Month

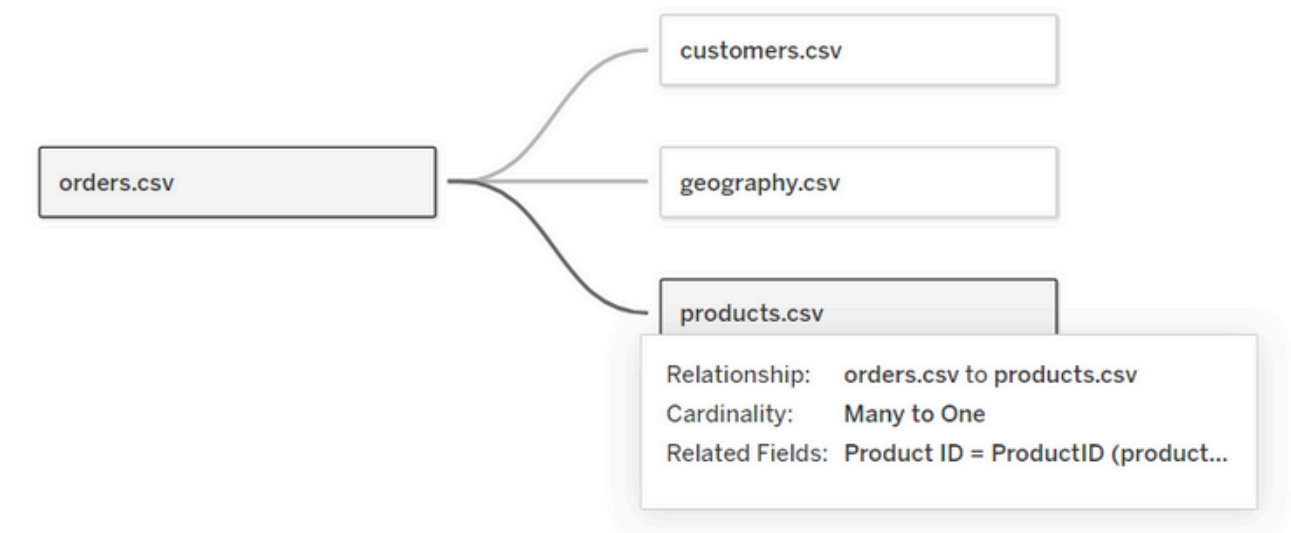
ESSENTIAL RELATIONSHIPS:

- Orders ↔ Products (Product ID)
- Orders ↔ Customers (Customer ID)
- Orders ↔ Geography (Region Code)

DATA TYPE CONSIDERATIONS:

- Dates: Ensure proper date formatting
- Numbers: Distinguish between dimensions and measures
- Text: Clean and standardise categorical data
- Geographic: Validate location data for mapping

orders+



KEY PERFORMANCE INDICATORS (KPI'S)



Revenue Metrics:

- **Total Revenue:** £2,347,890 (↑12% vs last month)
- **Average Order Value:** £84.50 (↓3% vs last month)
- **Monthly Recurring Revenue:** £195,658 (↑8% vs last month)



Customer Metrics:

- **Customer Acquisition Cost:** £15.20 (↑8% vs last month)
- **Customer Lifetime Value:** £425.80 (↑15% vs last month)
- **Monthly Active Customers:** 1,847 (↑22% vs last month)



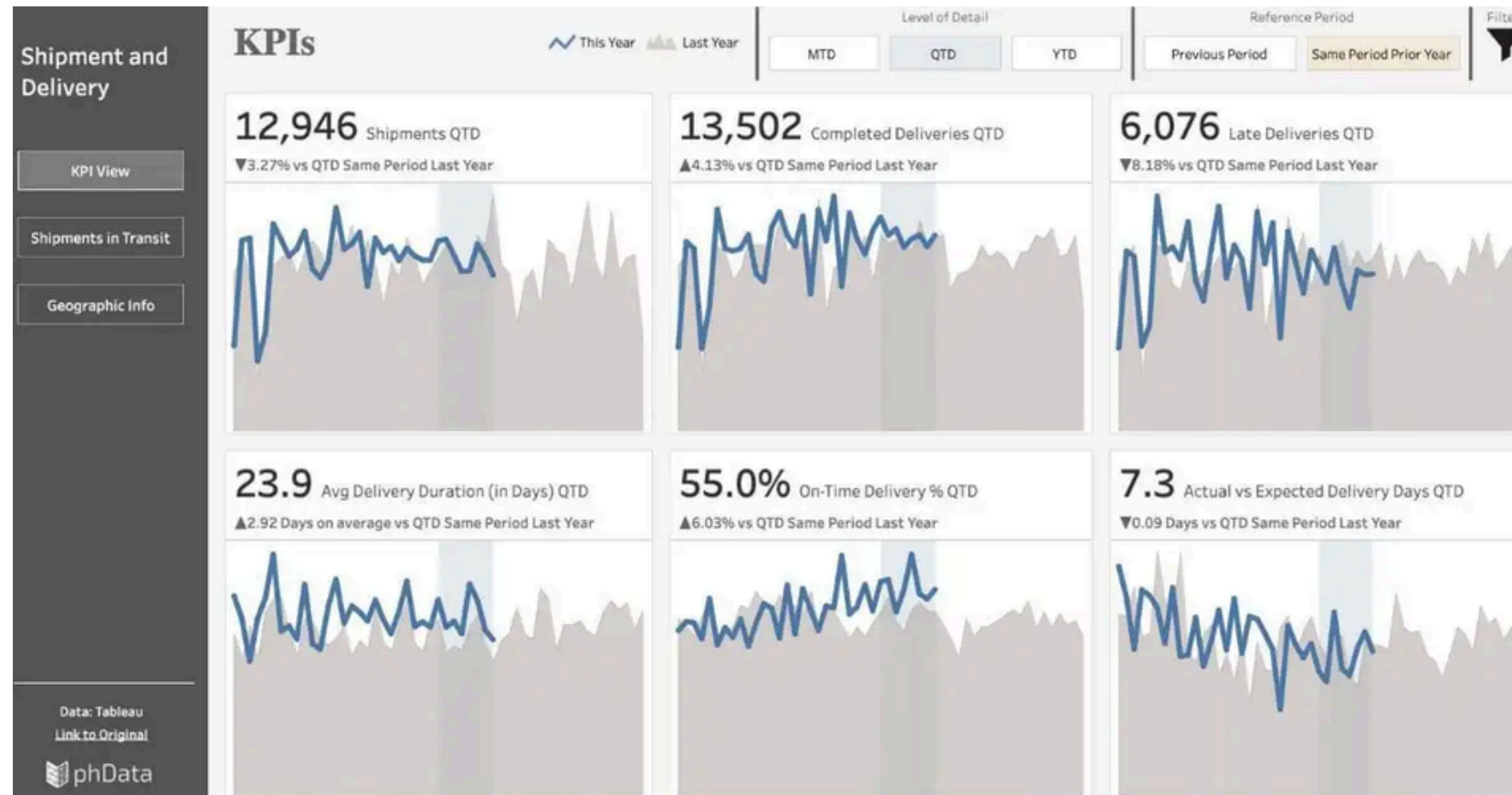
Operational Metrics:

- **Conversion Rate:** 3.2% (↑0.4% vs last month)
- **Cart Abandonment Rate:** 67.8% (↓2.1% vs last month)
- **Return Rate:** 8.5% (↑1.2% vs last month)

Why These Matter:

- Revenue metrics show business health
- Customer metrics guide acquisition and retention
- Operational metrics identify improvement opportunities

TIME SERIES ANALYSIS



Monthly Revenue Pattern:

- Peak Season: November-December (£400k+ monthly)
- Steady Growth: January-March (£180k-220k monthly)
- Summer Dip: July-August (£160k-180k monthly)
- Back-to-School: September boost (£250k)

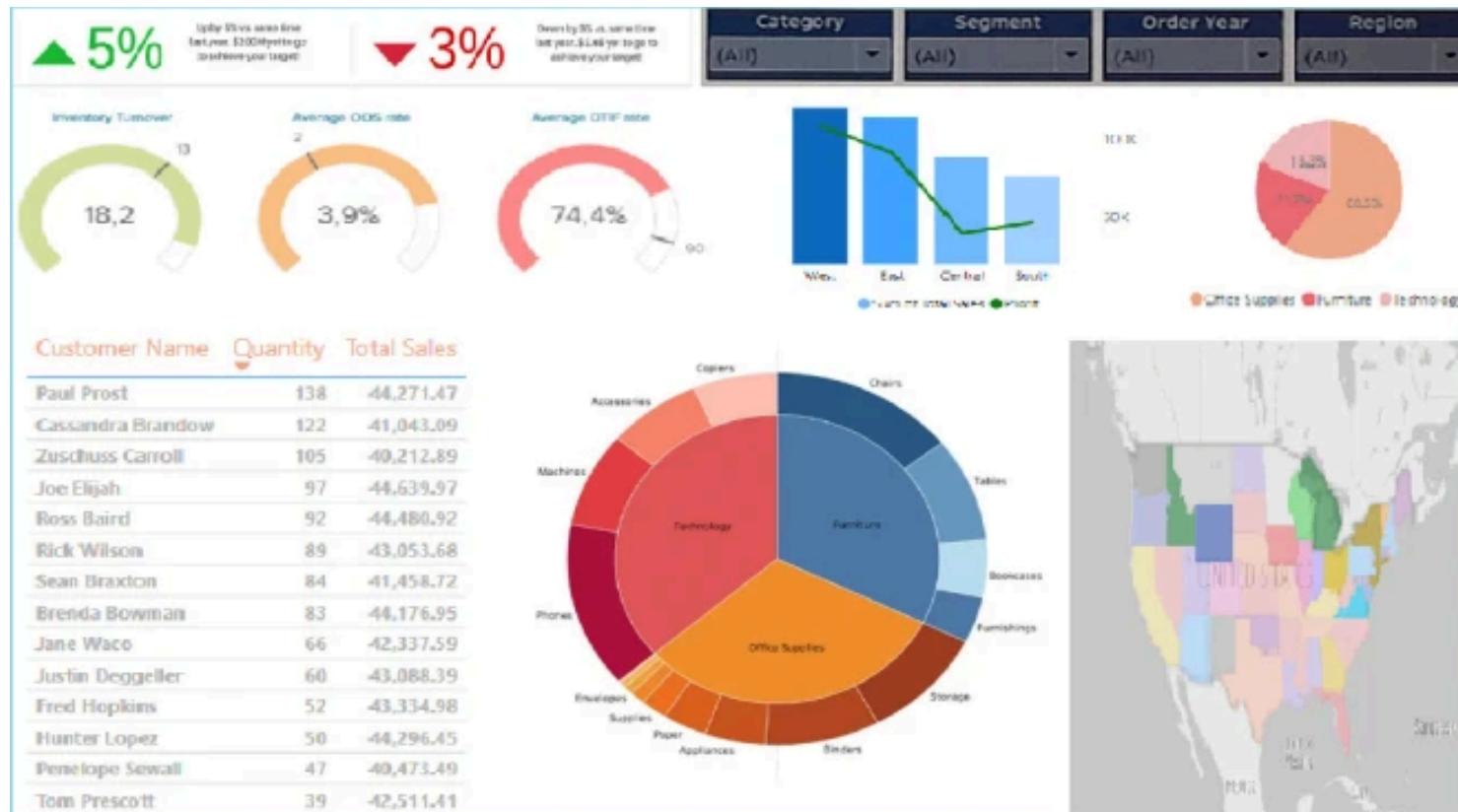
Key Insights:

- 45% of annual revenue occurs in Q4
- Consistent year-over-year growth of 18%
- Seasonal patterns are predictable and actionable
- COVID-19 impact visible in March-May 2020

Forecasting Implications:

- Inventory planning for seasonal peaks
- Marketing budget allocation by month
- Cash flow management for low periods
- Staff scheduling for peak seasons

PRODUCT CATEGORY ANALYSIS



Top Performing Categories:

- Electronics: £687k revenue (29% of total)
- Clothing: £562k revenue (24% of total)
- Home & Garden: £423k revenue (18% of total)
- Books: £298k revenue (13% of total)

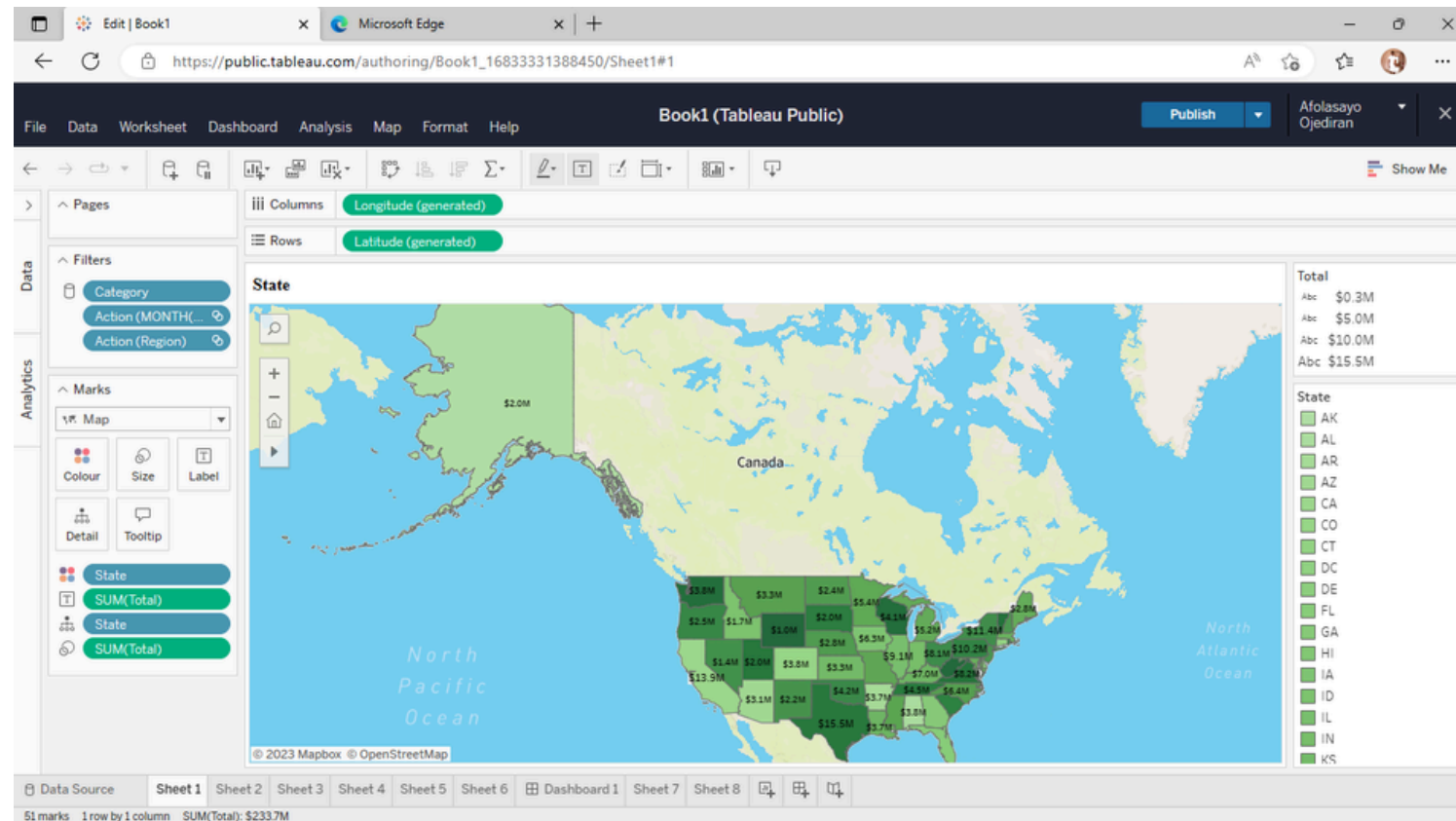
Category Insights:

- Electronics has highest revenue but lowest profit margins (12%)
- Clothing shows consistent growth across all seasons
- Home & Garden peaks during spring months
- Books declining 5% year-over-year

Strategic Implications:

- Focus marketing spend on high-margin categories
- Seasonal inventory planning by category
- Consider expanding successful product lines
- Investigate book category decline

GEOGRAPHIC PERFORMANCE



Revenue by Region:

- London: £758k (32% of total)
- South East: £445k (19% of total)
- North West: £367k (16% of total)
- Scotland: £298k (13% of total)

Regional Insights:

- London dominates but may be saturated
- Strong performance in traditional commercial centres
- Untapped potential in Wales and Northern Ireland
- Rural areas show lower online adoption

Expansion Opportunities:

- Targeted marketing in underperforming regions
- Regional pricing strategies
- Local partnership opportunities
- Logistics optimisation for remote areas

INTERACTIVE DASHBOARD FILTERS

Essential Filter Types

- **Date Range:** Monthly, quarterly, yearly views
- **Product Category:** Focus on specific segments
- **Geographic Region:** Regional performance analysis
- **Customer Segment:** New vs returning customers

Filter Best Practices

- **Logical Grouping:** Related filters together
- **Clear Labels:** Intuitive naming conventions
- **Default States:** Sensible starting positions
- **Visual Feedback:** Show filter impacts immediately

User Experience Considerations

- Maintain context when filtering
- Provide "reset" functionality
- Show number of records affected
- Ensure mobile-friendly design

DASHBOARD DESIGN PRINCIPLES

LAYOUT STRATEGY

- Top Row: Key metrics and KPIs
- Middle Section: Primary charts and trends
- Bottom Area: Detailed breakdowns and filters
- Right Panel: Context and explanations

VISUAL HIERARCHY

- Primary: Most important metrics (largest, top-left)
- Secondary: Supporting charts (medium size)
- Tertiary: Detailed data and filters (smallest)

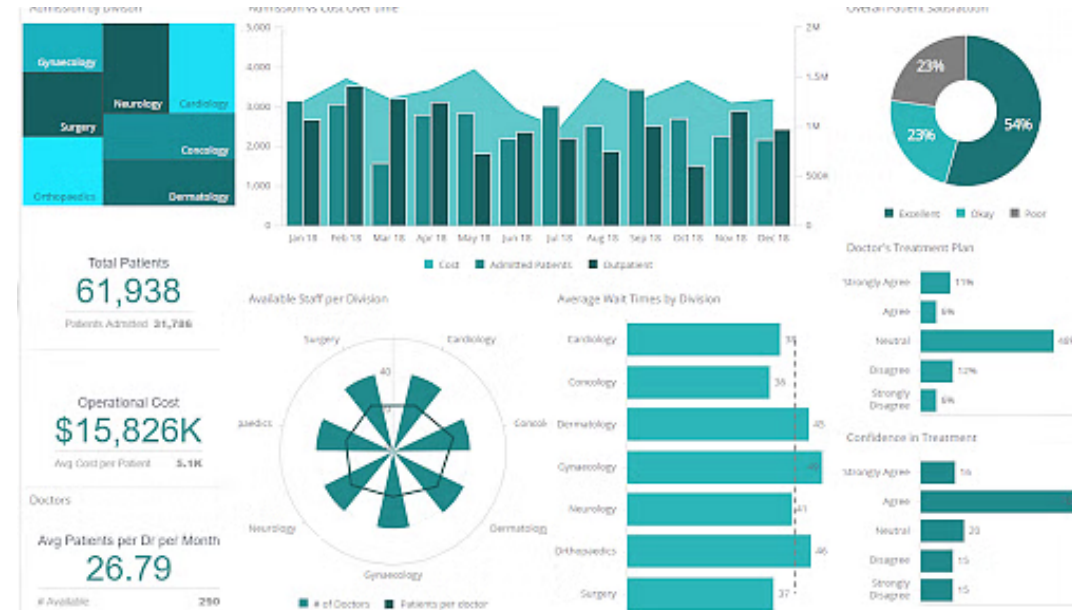
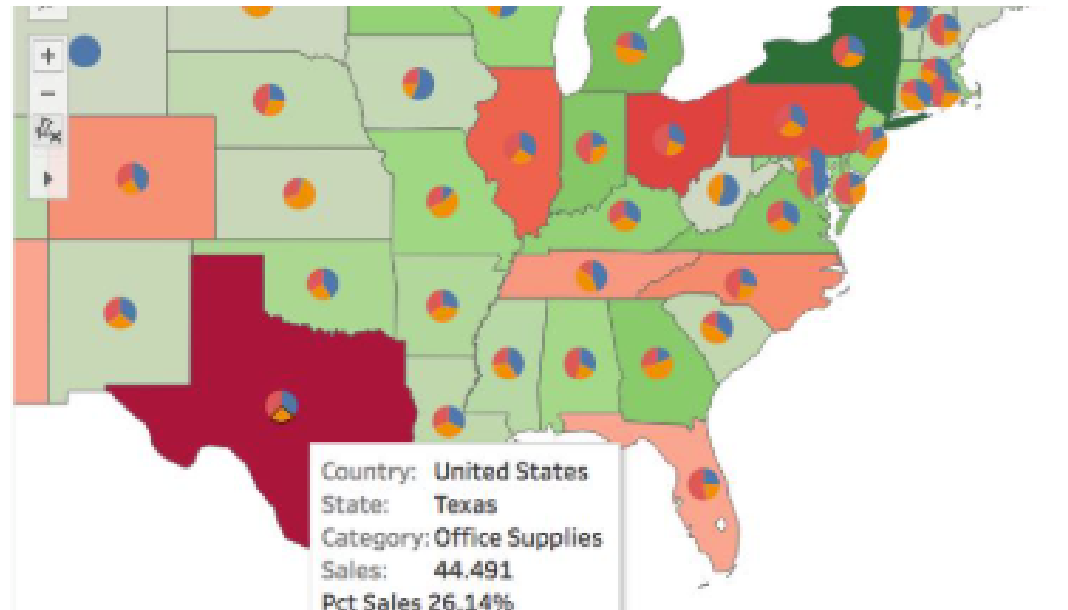
CONSISTENCY ELEMENTS

- Colour Palette: Maximum 5 colours with clear meaning
- Typography: 2-3 font sizes maximum
- Spacing: Regular grid system
- Alignment: Clean, professional appearance

USER TESTING

- Can users find key insights in 30 seconds?
- Do interactions work as expected?
- Is the story clear without explanation?

DRILL-THROUGH CAPABILITIES



Multi-Level Analysis

- **Level 1:** National performance overview
- **Level 2:** Regional breakdown and trends
- **Level 3:** City-level detail and opportunities
- **Level 4:** Individual customer behaviour

Navigation Design

- **Breadcrumbs:** Show current location in hierarchy
- **Click Indicators:** Visual cues for interactive elements
- **Context Preservation:** Maintain filters across levels
- **Return Options:** Easy navigation back to overview

Business Value

- **Executive Level:** High-level strategic insights
- **Management Level:** Operational performance detail
- **Analyst Level:** Detailed investigation capabilities
- **Field Level:** Local market intelligence

DASHBOARD BEST PRACTICES

Visual Design Principles:

- 5-Second Rule: Key insights visible immediately
- Colour Purpose: Use colour to highlight, not decorate
- White Space: Breathing room improves comprehension
- Consistency: Standardised fonts, colours, and layouts

DASHBOARD BEST PRACTICES

Content Strategy:

- Pyramid Structure: Most important information at top
- Context Always: Provide comparisons and benchmarks
- Actionable Insights: Clear next steps from data
- Story Flow: Logical progression through information

DASHBOARD BEST PRACTICES

Technical Considerations:

- Performance: Fast loading times essential
- Responsiveness: Works on all device sizes
- Accessibility: Colour-blind friendly palettes
- Scalability: Handles growing data volumes

COMMON MISTAKES TO AVOID

- 01 Too much information on one screen
- 02 Inconsistent colour coding
- 03 Missing context or benchmarks
- 04 Poor mobile experience

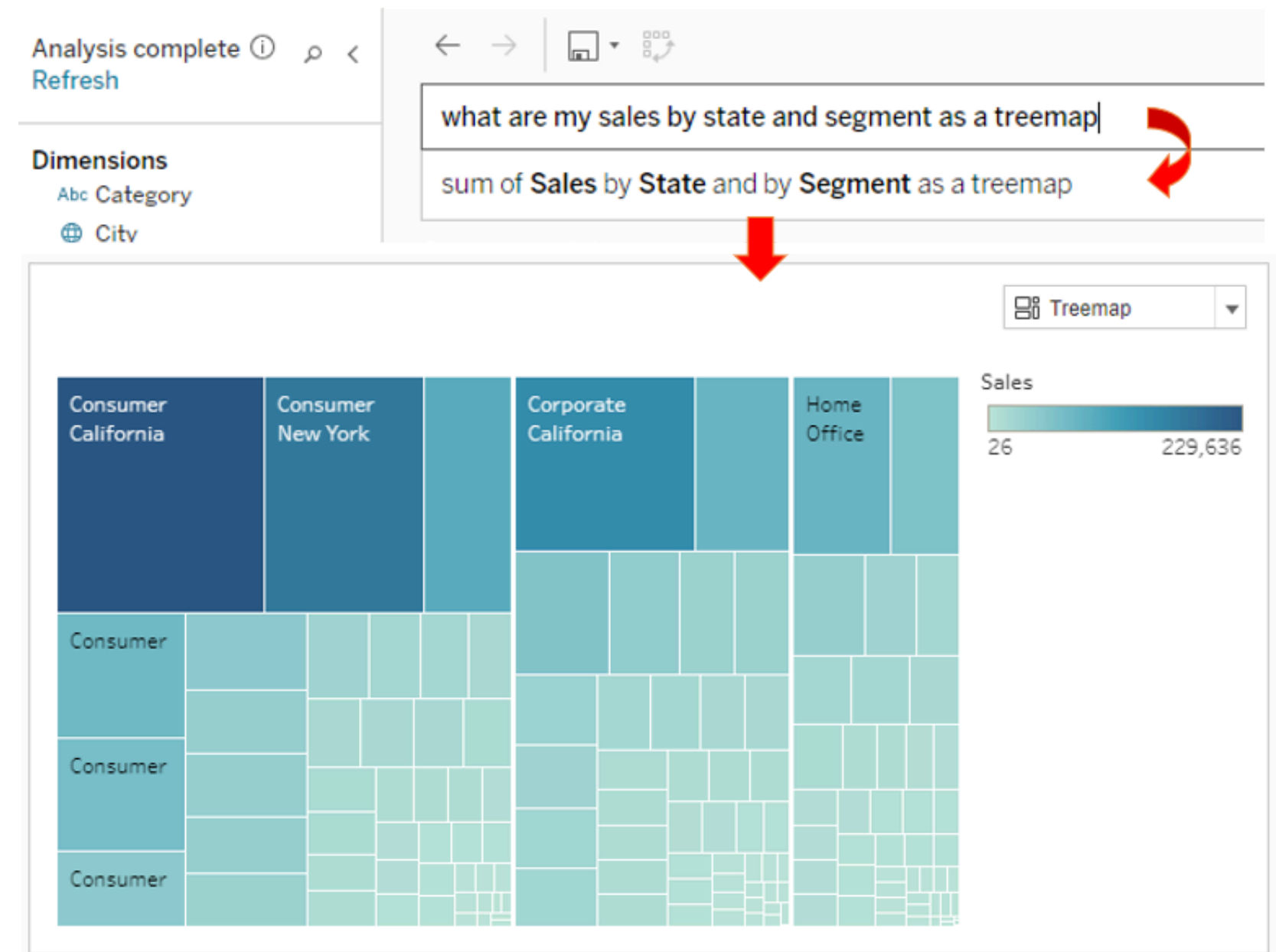
TABLEAU'S ASK DATA FEATURE

How Ask Data Works?

- Type questions in plain English
- Tableau interprets intent and generates visualisations
- Instant results with suggested follow-up questions
- Learns from your data structure and relationships

Example Queries:

- "Show me sales by region last quarter"
- "What products had the highest profit margin?"
- "How did marketing campaigns perform in 2023?"
- "Which customers haven't purchased recently?"



Benefits of Ask Data Feature

- **Accessibility:** No technical skills required
- **Speed:** Immediate insights without chart building
- **Exploration:** Encourages data discovery
- **Self-Service:** Reduces analyst workload

Limitations of Ask Data Feature

- **Simple Queries:** Complex analysis still requires traditional approach
- **Data Quality:** Depends on well-structured, clean data
- **Ambiguity:** May misinterpret unclear questions
- **Context:** Limited understanding of business nuances

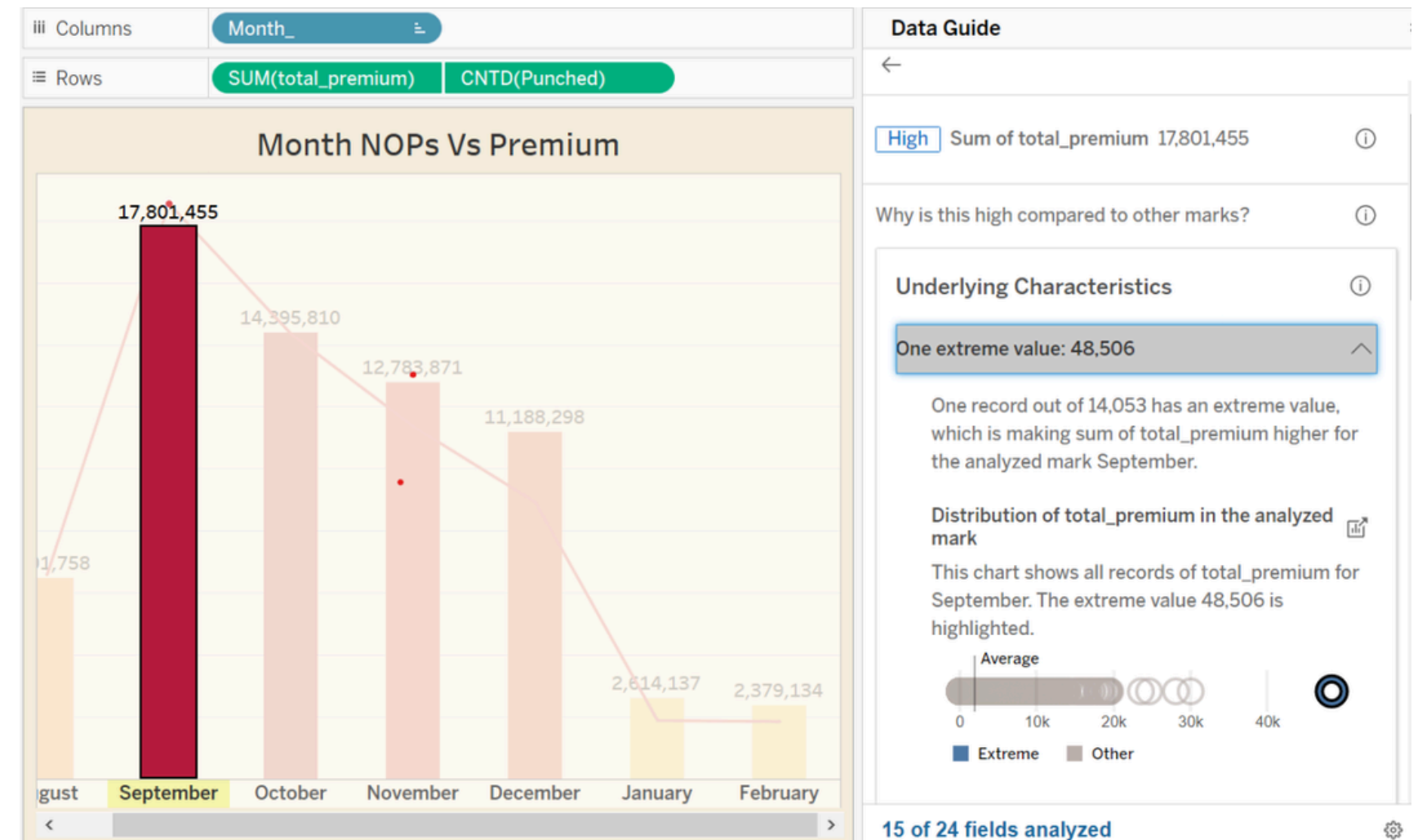
TABLEAU'S EXPLAIN DATA INSIGHTS

What Explain Data Does?

- Automatically identifies statistical outliers
- Suggests potential explanations for unusual patterns
- Provides confidence levels for insights
- Highlights correlations and trends

Types of Explanations:

- **Outlier Detection:** Unusual data points and their causes
- **Trend Analysis:** Underlying patterns in time series
- **Correlation Discovery:** Relationships between variables
- **Seasonal Patterns:** Recurring cycles in data



Business Applications

- **Anomaly Investigation:** Quickly understand unusual results
- **Root Cause Analysis:** Identify factors behind performance changes
- **Opportunity Identification:** Spot trends before they become obvious
- **Quality Assurance:** Detect data quality issues

Critical Considerations

- **Validation Required:** AI suggestions need human verification
- **Context Matters:** Statistical significance $\neq$ business relevance
- **Bias Awareness:** Explanations may reflect data biases
- **Domain Expertise:** Business knowledge essential for interpretation

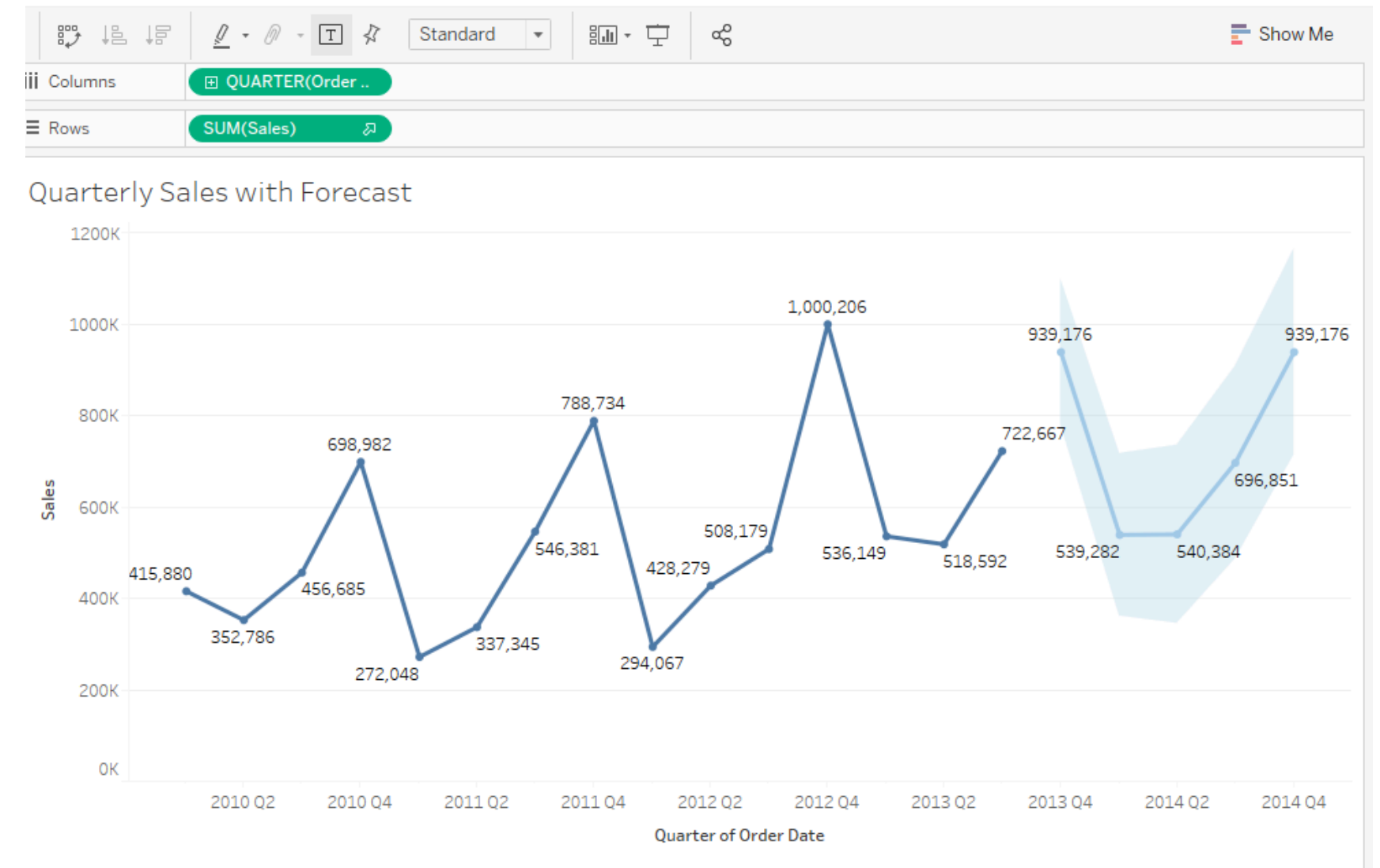
TABLEAU'S BUILT-IN FORECASTING

Forecasting Capabilities

- **Trend Detection:** Automatic identification of growth patterns
- **Seasonality:** Recognition of cyclical patterns
- **Confidence Intervals:** Uncertainty quantification
- **Multiple Models:** Exponential smoothing and other algorithms

Practical Applications:

- **Sales Forecasting:** Predict next quarter's revenue
- **Inventory Planning:** Anticipate demand fluctuations
- **Resource Allocation:** Staff scheduling based on predicted workload
- **Budget Planning:** Financial projections and variance analysis



Step by Step - Forecast in Tableau

~ <https://community.tableau.com/s/question/0D54T00000C6V3USAV/step-by-step-forecast-in-tableau>

Best Practices

- **Sufficient History:** Minimum 24 data points recommended
- **Data Quality:** Clean, consistent historical data essential
- **External Factors:** Consider events that may affect future performance
- **Regular Updates:** Refresh forecasts as new data arrives

Limitations

- **Black Swan Events:** Cannot predict unprecedented disruptions
- **Changing Patterns:** Struggles with structural shifts
- **External Variables:** Doesn't account for market changes
- **Accuracy Decreases:** Less reliable for longer time horizons

TABLEAU PREP OVERVIEW

Visual Data Preparation

Data Preparation Challenges:

- **Multiple Sources:** Combining data from different systems
- **Quality Issues:** Missing values, duplicates, inconsistencies
- **Format Variations:** Different date formats, naming conventions
- **Transformation Needs:** Calculations, aggregations, pivoting



Visual Interface: Drag-and-drop data transformation



Automated Suggestions: AI-powered cleaning recommendations



Flow Documentation: Clear audit trail of changes



Reusable Workflows: Save and share preparation processes

TABLEAU PREP OVERVIEW

Key Features:

- **Smart Defaults:** Automatic handling of common issues
- **Data Profiling:** Visual summary of data quality
- **Union and Join:** Combine multiple data sources
- **Pivot Operations:** Reshape data for analysis

Business Benefits:

- **Time Savings:** Reduce data preparation time by 60%
- **Consistency:** Standardised data processing
- **Collaboration:** Share preparation workflows
- **Governance:** Documented data lineage

BUSINESS PRESENTATION STRUCTURE

01 What

Present the Data

- Clear Findings: State your key insights directly
- Supporting Evidence: Show the data that backs up your claims
- Context: Provide benchmarks and comparisons
- Confidence Level: Indicate certainty in your conclusions

e.g.

Customer acquisition cost increased 23% in Q3

02 So What

Explain the Impact

- Business Implications: What does this mean for the organisation?
- Opportunity Size: Quantify the potential impact
- Risk Assessment: What happens if we don't act?
- Stakeholder Effects: Who is affected and how?

This threatens our profitability and growth targets

03 Now What

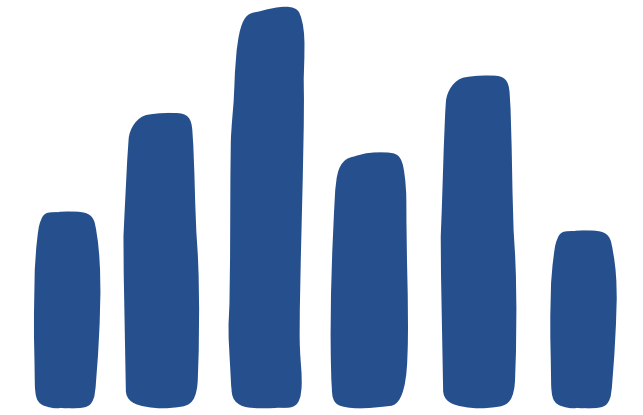
Recommend Actions

- Specific Steps: Clear, actionable recommendations
- Timeline: When actions should be taken
- Resource Requirements: What's needed to implement
- Success Metrics: How to measure progress

Optimise ad spend and improve conversion rates by December



THE ART OF DATA STORYTELLING



Story Structure:

- **Setup:** Establish **context** and business situation
- **Conflict:** Identify challenges or opportunities
- **Resolution:** Present insights and recommendations
- **Call to Action:** Specific next steps



Narrative Techniques:

- **Hook:** Start with a surprising or compelling fact
- **Progression:** Build complexity gradually
- **Tension:** Highlight stakes and **urgency**
- **Resolution:** Provide clear **solutions**



Visual Storytelling:

- **Consistent Design:** Maintain theme throughout
- **Progressive Disclosure:** Reveal information strategically
- **Emotional Connection:** Use visuals that resonate
- **Clear Pathway:** Guide audience through your logic

COMMON PITFALLS TO AVOID IN STORYTELLING

- **Data Dumping**
Showing everything instead of key insights
- **Lack of Context**
Not explaining why data matters
- **Weak Conclusion**
Failing to provide clear next steps
- **Overcomplication**
Making simple insights unnecessarily complex

QUESTIONS?